

GOVERNMENT CHALLENGE · SOLUTION PRESENTATION

SENTINEL VISION

Unified CCTV Intelligence & Real-Time Smart Policing Platform

Camera integration · AI video analytics · ANPR · Alerts & incidents · Evidence · GIS · Self-Heal

Every capability shown is implemented in the repository and labelled Current, Demonstrated or Future.

The problem control rooms face today

Fragmented CCTV

Cameras sit behind different vendors, recorders and VMS products, each with its own screen.

Manual monitoring burden

Operators watch many feeds at once; attention does not grow with camera count.

Delayed identification

A watch-listed vehicle or a restricted-area intrusion is often found after the fact.

Hard investigations

Tracing a vehicle means scrubbing footage camera by camera and collecting evidence by hand.

Camera health

Streams drop, credentials fail and bandwidth varies, often without anyone noticing.

SENTINEL VISION in one sentence

One web platform that connects cameras from any source, watches them with AI, raises alerts when rules are broken, and hands operators verifiable evidence and a map to act on.

Connect

One camera registry for webcam, video, RTSP and the Sentinel Grid

Detect

People, vehicles and number plates, per camera

Decide

Zones, watchlists and loitering rules raise alerts

Respond

Alerts grow into correlated incidents

Prove

Hashed snapshots, clips and incident evidence packages

Stay up

Self-Heal reconnects cameras and reports what failed

Solution architecture



Four tiers: Next.js dashboard · FastAPI REST + WebSocket · per-camera AI pipeline · SQLite / PostgreSQL + evidence store. Everything shown is implemented.

CCTV and camera integration

Central camera registry

CURRENT

Every camera in one catalogue with location, status and AI switches.

Camera adapters

CURRENT

Webcam, uploaded video, RTSP (TCP or UDP), mock VMS behind one interface.

Sentinel Grid

DEMONSTRATED

Catalogue sync of 30 cameras, autoconnect, bulk control. Streaming is partial today (see slide 12).

VMS federation

FUTURE

New VMS products join as adapters; ONVIF is an interface stub.

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Command Center

Live Cameras

AI Vision

Search

Alerts

Incidents

Investigate

Map

Evidence

Analytics

ADMINISTRATION

Cameras

Camera Control

Watchlists

AI Rules

Users

Audit

System

SELF-HEAL

Health Dashboard

Problems

Recovery Activity

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System online

System Administrator

Logout

Search camera, person, vehicle, plate...

Camera Control Center

Bulk and individual control over every registered camera.

Connect All

Start All

Start AI All

Restart All

Stop AI All

Disconnect All

CAMERA	LOCATION	CONNECTION	AI	FPS	LATENCY	LAST ERROR	ACTIONS
GRID-cam01	01 Chiman bhai Bridge	online	AI RUNNING	30	—	—	⋮
GRID-cam02	02 Janpath	online	AI RUNNING	30	—	—	⋮
GRID-cam03	03 O.N.G.C. Office	online	AI RUNNING	30	—	—	⋮
GRID-cam04	04 Paldi Circle	online	AI RUNNING	250	—	—	⋮
GRID-cam05	05 Visat teen Rasta	online	AI RUNNING	30	—	—	⋮
GRID-cam06	06 Timbavadi gate-Junagadh	online	AI RUNNING	25	—	—	⋮
GRID-cam07	07 hero-showroom-gir-somnath	online	AI RUNNING	25	—	—	⋮
GRID-cam08	08 majewadi-gate-junagadh	online	AI RUNNING	25	—	—	⋮
GRID-cam09	09 new-bypass-near-by-circle-junagadh-2	online	AI RUNNING	25	—	—	⋮
GRID-cam10	10 char-chowk-road-2-junagadh	online	AI RUNNING	25	—	—	⋮
GRID-cam11	11 dolatpara-junagadh	offline	—	25	—	—	⋮
GRID-cam12	12 Tri Mandir Adalaj Tollnaka	online	AI RUNNING	20	—	—	⋮
GRID-cam13	13 CN Vidhyalaya	online	AI RUNNING	10	—	—	⋮
GRID-cam14	14 Delight RLVD	online	AI RUNNING	10	—	—	⋮
GRID-cam15	15 Suvidha park	offline	AI STARTING	10	—	—	⋮

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AI video analytics

Ultralytics YOLO11s

CURRENT

Person and vehicle detection at 960 px on one shared GPU model (successor to the earlier YOLOv8s baseline).

ByteTrack

CURRENT

A tracker per camera, so every object keeps one ID across frames.

EasyOCR ANPR

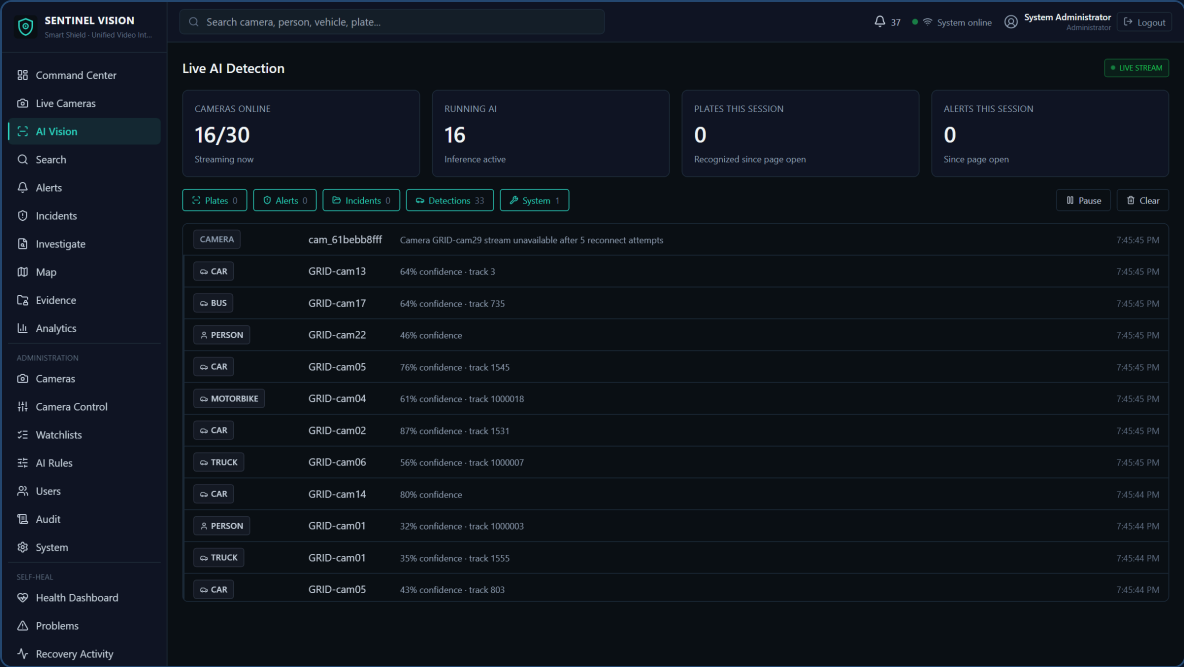
CURRENT

A plate detector finds the plate; EasyOCR reads it; reads are voted across frames.

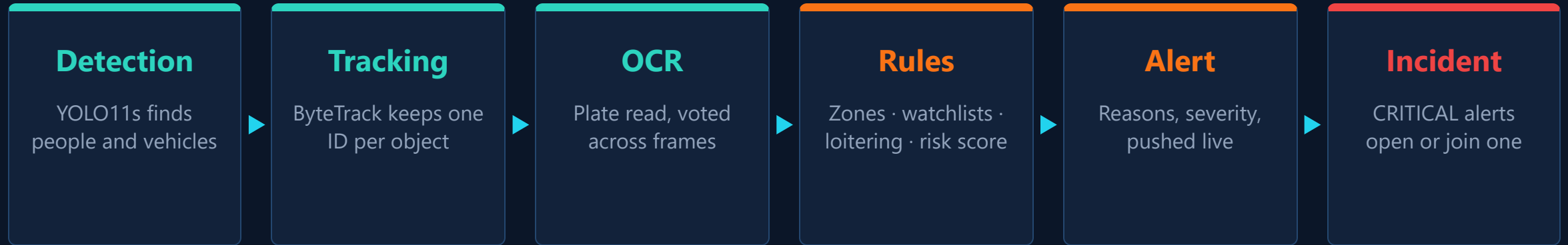
Selective analytics

CURRENT

Person, vehicle and ANPR switched per camera; AI slots rotate when hardware is short.



Intelligent event processing



A one-frame plate read is stored but goes to review; CRITICAL watchlist alerts need corroboration across frames.

Cooldowns per camera, rule and track stop one object raising the same alert every frame.

One vehicle crossing several cameras becomes one incident whose title and priority grow with it.

GIS and camera intelligence

19 of 30 grid cameras mapped

CURRENT

Positions looked up in OpenStreetMap from each camera's place name.

11 intentionally not mapped

LIMITATION

No reliable match, so they are not guessed; the map says so.

Clustered markers + legend

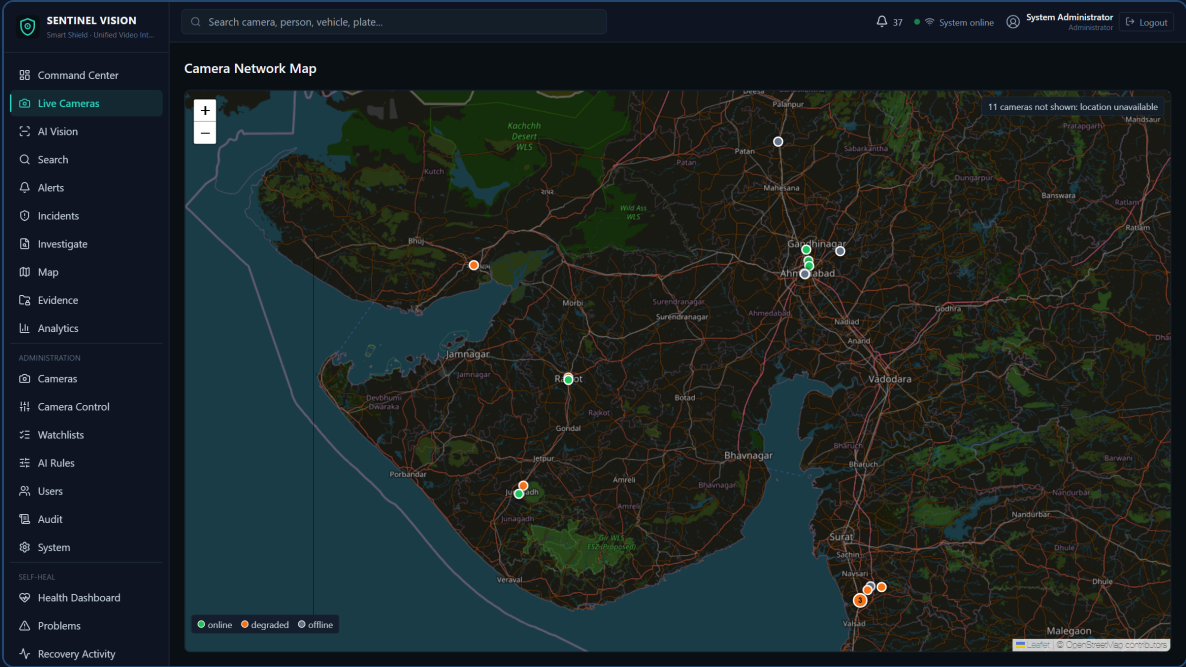
CURRENT

Cameras sharing a spot (three in Bilimora) form one marker; amber = degraded.

Vehicle journey map

CURRENT

Sightings drawn on camera positions; GRID-cam06 verified at Timbavadi, Junagadh.



Alerts, incidents and evidence

Alerts

CURRENT

Reasons, severity and risk factors; acknowledge, escalate or dismiss.

Incidents

CURRENT

Correlated across cameras, with notes and a summary.

Evidence

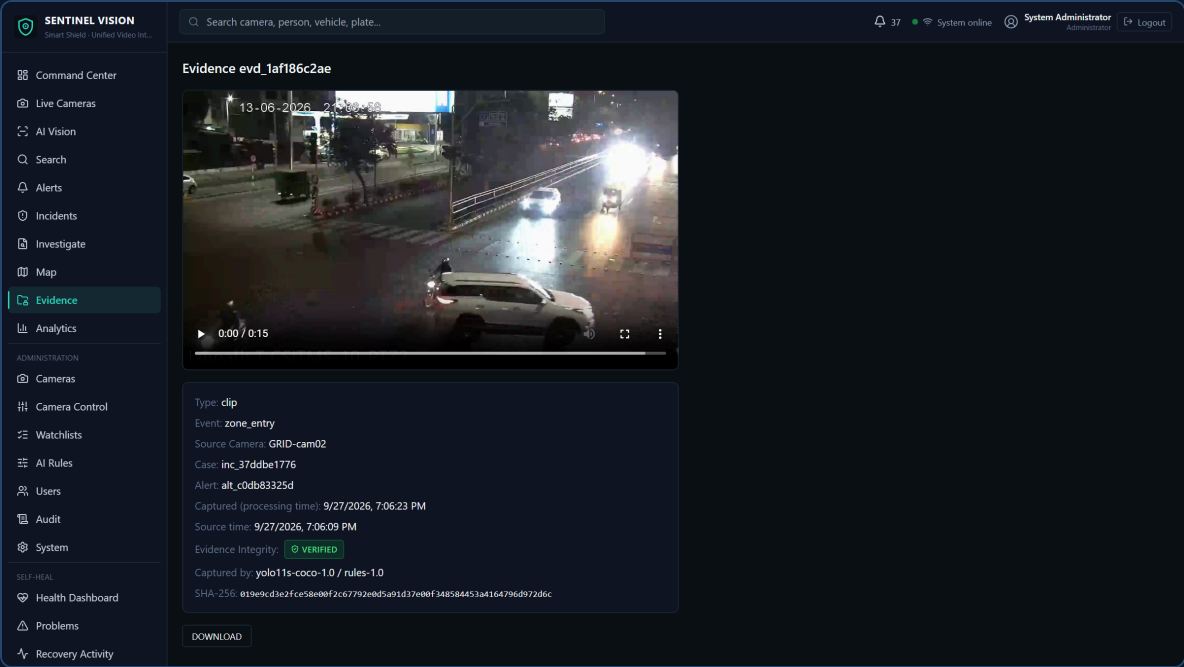
DEMONSTRATED

Snapshot per alert, event clips, operator REC recordings; SHA-256 at capture, VERIFIED on access.

Evidence package

CURRENT

Per incident, JSON or PDF, optional plate redaction; every access audited.



Self-Heal and reliability

Camera health

CURRENT

Each camera reports CONNECTED, PROCESSING, DEGRADED, RECONNECTING or AUTH_ERROR.

Reconnect + backoff

CURRENT

Per-camera backoff; a grid-wide breaker pauses 5 to 60 min and probes with one camera first.

Database retry

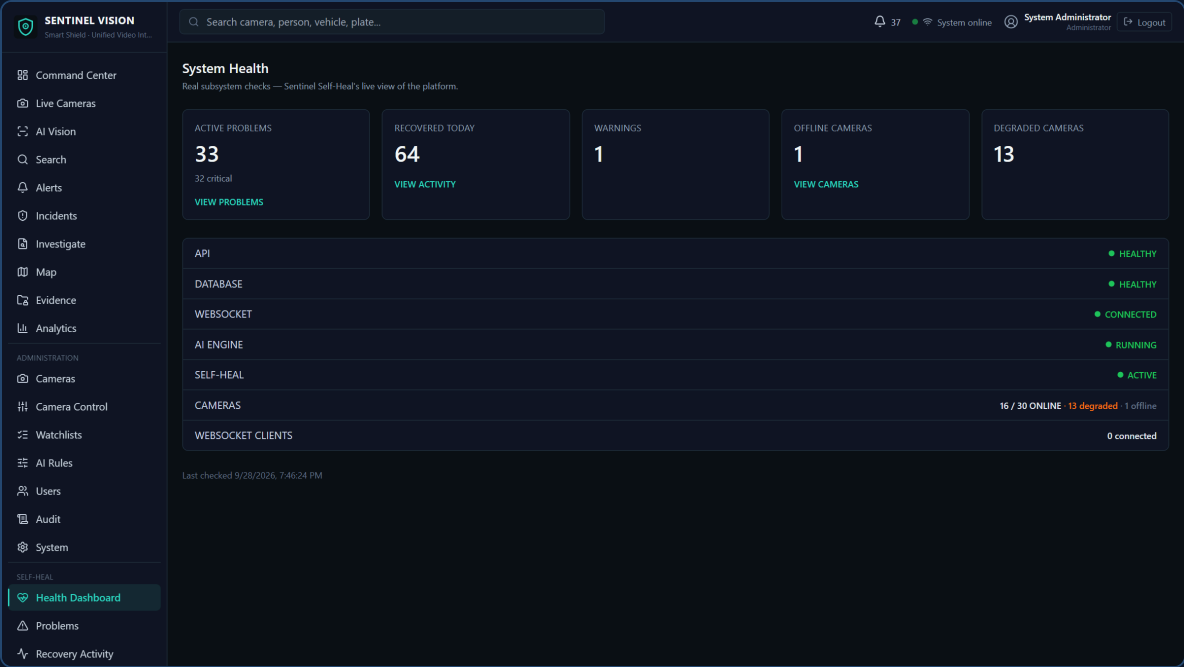
CURRENT

Locked writes are rolled back, reapplied and logged as Self-Heal events.

Failure isolation

CURRENT

One camera crashing never stops the others; the dashboard WebSocket reconnects itself.



Security by design

Authentication

CURRENT

bcrypt password hashes, JWT sessions, login rate limiting.

RBAC

CURRENT

Five roles checked on every API route: Administrator, Control Room Operator, Investigator, Supervisor, Auditor.

Resource tokens

CURRENT

Short-lived signed tokens for live streams (1 h) and evidence files (5 min).

Audit trail

CURRENT

Hash-chained audit log; a verify endpoint detects any edited or deleted row.

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Command Center

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SELF-HEAL

Health Dashboard

Problems

Recovery Activity

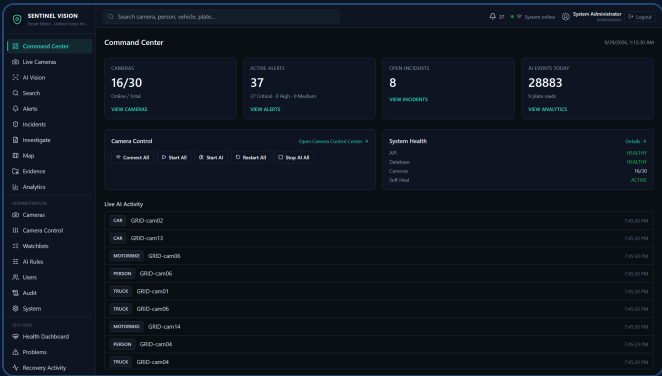
Search camera, person, vehicle, plate...

Audit Logs

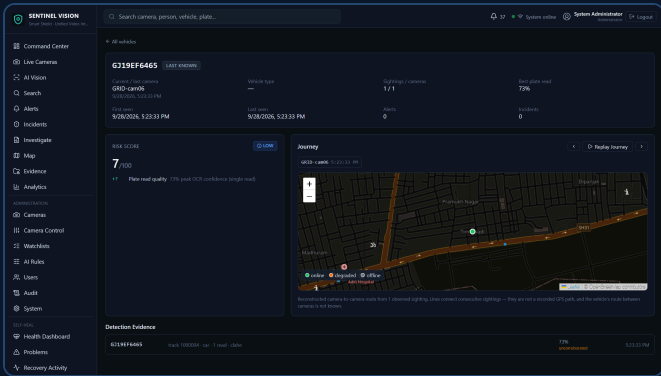
Filter by user...FILTER

TIMESTAMP	USER	ACTION	RESOURCE	RESULT
9/28/2026, 7:46:06 PM	admin	download_evidence	evd_1af186c2ae	SUCCESS
9/28/2026, 7:46:05 PM	admin	request_evidence_file_token	evd_1af186c2ae	SUCCESS
9/28/2026, 7:45:27 PM	admin	login		SUCCESS
9/28/2026, 7:45:24 PM	admin	login		SUCCESS
9/28/2026, 7:41:53 PM	admin	login		SUCCESS
9/28/2026, 7:12:51 PM	admin	login		SUCCESS
9/28/2026, 7:12:41 PM	admin	login		SUCCESS
9/28/2026, 7:12:31 PM	admin	login		SUCCESS
9/28/2026, 7:12:20 PM	admin	login		SUCCESS
9/28/2026, 7:12:10 PM	admin	login		SUCCESS
9/28/2026, 7:12:00 PM	admin	login		SUCCESS
9/28/2026, 7:11:49 PM	admin	login		SUCCESS
9/28/2026, 7:11:39 PM	admin	login		SUCCESS
9/28/2026, 7:11:29 PM	admin	login		SUCCESS
9/28/2026, 7:11:18 PM	admin	login		SUCCESS
9/28/2026, 7:11:08 PM	admin	login		SUCCESS
9/28/2026, 7:10:58 PM	admin	login		SUCCESS
9/28/2026, 7:10:48 PM	admin	login		SUCCESS
9/28/2026, 7:10:37 PM	admin	login		SUCCESS
9/28/2026, 7:10:27 PM	admin	login		SUCCESS

What works today



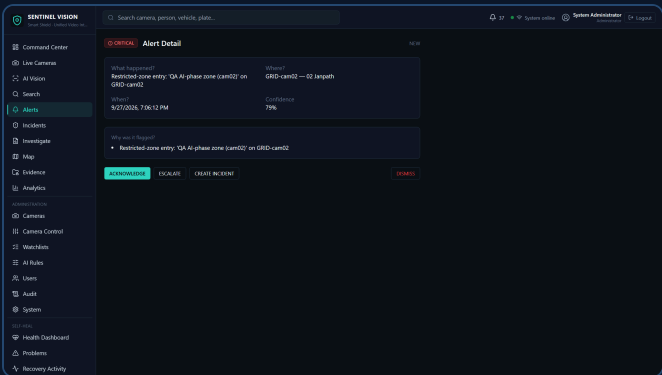
Command centre



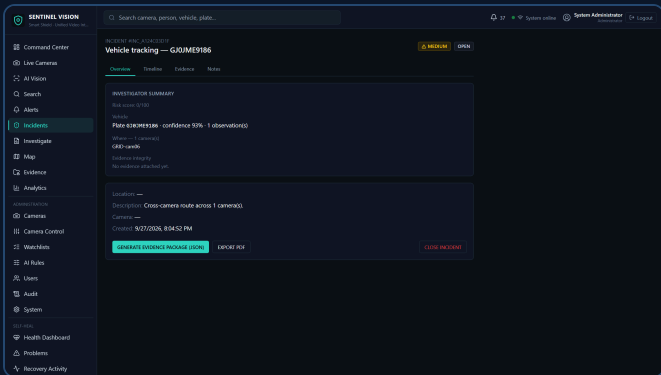
Vehicle journey

Demonstrated

Login, dashboard, camera registry and map, live AI detection and tracking, ANPR, zone alerts, incidents, evidence with integrity check, Self-Heal screens.



Alert detail



Incident

Known limits

Grid streaming is partial: about 29 cameras degraded and about 3.4 Mbit/s observed recently. SQLite is single-writer; a login-contention fix is tested but not yet re-verified with 30 live grid streams.

From today to larger deployments

Current implementation

- Single backend process, one asyncio task per camera
- SQLite (WAL) for the demo
- PostgreSQL 16 + PostGIS in Docker Compose, built and migrated by CI
- Valkey for shared cooldowns and rate limits (optional)
- Tested: 1101 backend tests, 23 frontend unit tests, 10 browser tests

Planned / future scale

- PostgreSQL as the default database, load-tested with a live fleet
- Separate AI worker processes / nodes per camera group
- Event streaming between ingestion, analytics and alerting
- Edge AI at camera clusters sending events, not full video
- More VMS adapters (ONVIF, vendor SDKs); container orchestration

Expected benefits and how to measure them

Expected benefits

- Operators work from alerts instead of watching every feed
- Faster identification of watch-listed vehicles and zone intrusions
- Investigations start from a vehicle journey and an evidence package
- Camera failures are visible and mostly self-recovering
- Every action and evidence access is accountable

Evaluation metrics for a pilot

- Time from event to operator alert
- False alerts per camera-hour
- Plate read rate and plate accuracy on the pilot's own cameras
- Camera uptime and time to recover
- Time to assemble an incident's evidence

SENTINEL VISION

A working, tested CCTV intelligence platform: cameras from mixed sources, AI detection, tracking and plate reading, rule-based alerts and incidents, verifiable evidence, a camera map and Self-Heal, secured with RBAC and a tamper-evident audit trail.

Built and demonstrated on real footage from a 30-camera grid

Honest about limits: partial grid streaming, SQLite demo database, 19 of 30 cameras mapped

Clear next step: a pilot on PostgreSQL with separate AI workers

Thank you · github.com/RuudyLinux/SENTINEL-AI